LEAD-BASED PAINT INSPECTION

At

ALBUQUERQUE SERVICE UNIT
Building No. 21

for

Albuquerque Area Indian Health Service
Division of Health Facilities
5300 Homestead Road, NE
Albuquerque, New Mexico 87110

By

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Federal Occupational Health
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1.0 INTRODUCTION

A Lead-Based Paint (LBP) Inspection was conducted in Building No. 21 at the Indian Health Service (IHS), Albuquerque Service Unit, Albuquerque, New Mexico on April 14, 2006. The purpose of the inspection was to identify lead in paint on interior and exterior painted surfaces. Measurements were made using a Niton X-ray Fluorescence (XRF) Spectrum Analyzer. Confirmatory chip sampling for laboratory analysis proved to be unnecessary. Mr. Alex Fontenarosa, Lead Inspector/Risk Assessor, representing Federal Occupational Health (FOH), conducted the survey.

Sampling procedures similar to the U.S. Department of Housing and Urban Development (HUD) *Guidelines for the Evaluation and Control of Lead-Based Paint Hazards in Housing*, dated June 1995, and revised in 1997, were followed for Building No. 21. Within each area sampled, procedures similar to the HUD multifamily housing inspection, as outlined in Chapter 7 of the HUD Guidelines, were employed.

Building 21 is a four-story building, with a penthouse, which is currently being used as a clinic. It appeared to be constructed two phases. The interior is approximately 59,867 square feet.

2.0 SURVEY METHODS

2.1 XRF Instrumentation

Measurements of lead in paint were made using a Niton XRF Spectrum Analyzer (serial number XL309-U3978NR6817). The XRF Lead Paint Analyzer non-destructively measures lead concentrations of painted surfaces. The XRF system includes three main components: (1) the XRF analyzer, which is a portable analytical instrument operating on the principles of X-ray fluorescence; (2) forms for recording of field data; and (3) report generation software, which allows for generation of data tables.

The criteria for classifying results obtained using the XRF are as shown in Table 1 below. These results are taken from the HUD/EPA Performance Characteristics Sheet, specific for the XRF system used in this survey.

Table 1:
LBP Classification Criteria, K-Shell Value

Substrate	Negative	Inconclusive	Positive
Brick	≤ 0.9	None	≥ 1.0
Concrete	≤ 0.9	None	≥ 1.0
Drywall	≤ 0.9	None	≥ 1.0
Metal	≤ 0.9	None	≥ 1.0
Plaster	≤ 0.9	None	≥ 1.0
Wood	≤ 0.9	None	≥ 1.0

*Values Reported as Milligrams per Square Centimeter (mg/cm^2)

FOH Lead Inspectors are licensed by the State of Utah Department of Environmental Quality, Division of Radiation Control, to own and operate x-ray fluorescence analyzers in Utah. FOH representatives obtained a Reciprocity Agreement with the State of New Mexico Radiation Regulatory Agency, which allows operation of XRF instruments in New Mexico under the Utah license. Mr. Fontenarosa is certified by the United States Environmental Protection Agency (EPA) to perform LBP inspections. Copies of certifications for Mr. Fontenarosa can be found in Appendix C.

2.2 XRF Calibration Check

Standard quality control procedures with respect to the XRF were performed during the survey. The internal calibration of the XRF was checked by taking a minimum of three consecutive measurements on a National Institute for Standards and Technology (NIST) standard with a known concentration of lead. The results of the calibration checks were then used to verify that the instrument was operating within acceptable limits. These calibration checks are reported within the XRF data tables found in Appendix A of this report and are maintained in a file to detect changes in instrument performance over time.

2.3 Substrate Correction

For some XRF instruments it is necessary to make corrections for bias in XRF readings made on some painted surfaces or "substrates". In accordance with the HUD/EPA Performance Characteristics Sheet for the XRF analyzer used, it is not necessary to correct for substrate bias.

2.4 Field Documentation

Field data forms were used to manually record information about individual XRF measurements. This information included the room number, the building component (such as wall, ceiling, door, baseboard, cabinet, etc.), component substrate (for example drywall, wood, metal, concrete), and paint condition. The field data forms were then used by the inspector to couple measurement-specific descriptive information to the sequential data once the instrument's electronic memory was downloaded to a personal computer. This information was then manipulated by the instrument's software to generate data tables for the written report.

According to the HUD Guidelines, each enclosed space in a structure (e.g., room, stairway, hallway, etc.) is considered a separate "room equivalent" and must be tested separately. To keep track of where individual measurement locations were made, interior and exterior walls of a structure were designated as A, B, C, and D, as shown in the attached drawings. The A wall was designated as the North wall in this inspection, and the B, C, and D walls are in clockwise order from the A wall. The wall and position on the wall (i.e., left, right, or center) was also recorded when readings were taken on components such as windows or doors. Any comments regarding a particular room were also recorded and may be found in the data tables.

Paint color and condition are noted and included in the data tables. Under Paint Condition an "I" indicates intact or sound paint; an "F" indicates fair condition (paint surfaces show minor damage); a "P" indicates poor condition (paint is peeling, cracking, or blistering or shows other significant damage).

The report generation software combined the lead readings stored in the XRF analyzer with the test sample location information recorded in the field to produce data tables. The transfer of data from the analyzer to the computer was made through an interface cable supplied as part of the XRF system.

3.0 LEAD-BASED PAINT DEFINITIONS

The U.S. Department of Housing and Urban Development (HUD) considers paint to be lead-containing if the concentration is 1.0 milligram of lead per square centimeter (1.0 mg/cm^2) or greater, or if the lead concentration is greater than 0.5% by weight. The Consumer Product Safety Commission (CPSC) currently considers paint to be lead-containing if the concentration of lead exceeds 600 ppm (0.06% by weight). In 1978, the CPSC banned the sale of lead-based paint to consumers and its application to areas where consumers have direct access to painted surfaces. This survey classified LBP on the basis of the HUD 1.0 mg/cm^2 criteria and 0.5% by weight (5000 ppm).

By contrast, the mission of the Occupational Safety and Health Administration (OSHA), with respect to lead-containing paint, is to protect workers during construction activities that may generate elevated airborne lead concentrations. OSHA states that construction work (including renovation, maintenance, and demolition) carried-out on structures coated with paints that have lead concentrations lower than the HUD or CPSC can still result in airborne lead concentrations in excess of regulatory limits. For this reason, OSHA has not defined lead-containing paint, but states that paint having any measurable level of lead may pose a substantial exposure hazard during construction work, depending upon the work performed. The owner or employers must be aware of the OSHA Lead in Construction Standard (29 CFR 1926.62) and the Respiratory Protection Standard (29 CFR 1910.134).

4.0 RESULTS SUMMARY

Lead paint inspection data tables are located in Appendix A of this report. The tables show all lead measurements in the order taken for each of the structures and sampling areas tested. Descriptions of the columns found in the data tables are described in Appendix A.

According to HUD Guidelines, lead-based paint was identified at or above 1.0 mg/cm^2 on interior and exterior building components sampled in the 1930 section of the hospital. No lead based paint was found in the 1980 addition to the hospital. The following list identifies room number, building component, paint color and condition, and approximate square footage of lead-based paint materials.

Table 2:
Albuquerque Service Unit
Building No. 21
XRF Spectrum Analyzer Results
Sampled on: April 14, 2006

Room No.	Building Component	Color/Condition	Approx. ft^2
Interior -1930 Section Only	Metal Doorframe Components	All/Intact	2,700ft^2
Interior -1930 Penthouse Only	Wood Door Components	All/Fair	120 ft^2
Interior -1930 Section Only	Wood Doors (X-Ray area only)	Varnish/Intact	135 ft^2
Interior -1930 Section Only	Metal Window Components	White/Intact	1,350 ft^2
Exterior -1930 Section Only	Metal Window Components	All/Fair	1,350 ft^2
Interior -1930 Section Only	Plaster Walls and Ceilings	All/Intact	70,363 ft^2
Interior -1930 Section Only	Metal Fire Boxes	All/Intact	24 ft^2
Exterior	Concrete Curb	Blue/Intact	200 ft^2
Exterior	Asphalt Stripes	White/Intact	500 ft^2
Interior– Stairwells– 1930 Section Only	Metal Stair Components	All/Intact	2,400 ft^2
Exterior	Metal Fire Hydrant	Yellow/Intact	20 ft^2
Room 503	Concrete Walls A and C	White/Blue/ Intact	400 ft^2
Room 503	Concrete Window Sill	White/Intact	5 ft^2
Exterior	Metal Cage	Red/Intact	30 ft^2
Room 119	Brick Wall A and B	White/Intact	410 ft^2
Rooms 410, 418, 313B, and Hallways	Drywall Walls	White/Intact	6,000 ft^2
Room S301	Concrete Walls B and C	White/Intact	800 ft^2

5.0 DISCUSSION AND RECOMMENDATIONS

Lead-based paint (LBP) was detected on interior and exterior building components in the 1930 section of the hospital during sampling conducted by FOH on April 14, 2006. No lead was found in the 1980 addition to the hospital. The majority of the interior LBP components were in good condition. Due to the observed good condition of the LBP components, FOH does not recommend abatement or maintenance attention to the LBP unless they will be disturbed by renovation or demolition activities. Several exterior LBP were in poor condition and do require maintenance attention

The mere presence of LBP in a building poses no health hazard so long as the paint remains intact. Lead is only a hazard to adults and workers when it is inhaled, as fine dust or fume, or when it is ingested, when a painted surface is mouthed or chewed by a child, or inadvertently through contamination of surfaces, hands, and food by lead-contaminated dust. Therefore, to minimize the potential hazards associated with the presence of LBP, the following procedures should be implemented.

- Identification (through testing) of all known or suspect LBP surfaces and communication of the presence of LBP surfaces to workers potentially affected by the presence of LBP. Although the LBP testing has been completed, completion of a specific Operations and Maintenance (O&M) Program is suggested to organize a coherent, continuous program outlining procedures for routine inspections, maintenance, and immediate clean-up of damaged LBP surfaces. The O&M Program should address periodic inspections, maintenance activities, methods of safe and prompt clean-up when a licensed lead abatement contractor is not used, work procedures which are likely to create lead hazards, and a program of standard work procedures which are designed to minimize lead contamination. Development of an O&M Program was not part of FOH's original Scope of Work.

The on-site maintenance should perform periodic inspections to assess the condition of LBP (i.e. chip, peel, or flake). If areas of LBP become damaged or deteriorated, a licensed lead abatement contractor should be contacted regarding removal or abatement. Maintenance activities which should be avoided in areas where known or suspect LBP is located include, but are not limited to:

- Scraping, sanding, or using a heat gun on painted surfaces before repainting;
- Making holes in walls to access at pipes, install electrical outlets, etc.;
- Removing building components for renovation activities;
- Repeatedly bumping furniture or other objects into painted components;
- Unnecessarily opening and closing windows with LBP components.

Major work or renovation performed on LBP components should be completed by a licensed LBP abatement contractor. However, if minor repair or immediate clean-up is necessary for a specific LBP component, at a minimum, the following work and clean-up procedures should be incorporated:

- Move children and pregnant women to another area of the facility until work is completed and the area is properly cleaned;
- Cover exposed areas using plastic to seal off entrances and ducts and protect furniture, carpets, rugs, and floors from paint dust and chips. Plastic should be carefully disposed;
- Wet the painted surface with soapy water prior to working;
- Thoroughly clean area following work. Clean-up dust and chips with wet mops or rags soaked in a solution of trisodium phosphate (TSP) or phosphate containing powdered dishwashing detergent. Use two buckets of water – one for wash and one for rinse. To prevent recontamination of cleaned surfaces, wash mops and rags thoroughly after each use. Dust, chips, debris, mops, and rags should be contained in plastic bags and disposed of properly by a licensed lead abatement contractor;
- Avoid sweeping or vacuuming the work area. Sweeping and vacuuming spread lead dust particles, since the filters are not fine enough to contain LBP particles;
- If an entire LBP component (such as a window sill or door frame) requires removal, a representative sample should be taken of the LBP-coated component and this sample should be analyzed via the Toxic Characteristic Leaching Process (TCLP) method to determine whether the component must be disposed of as a hazardous waste. A licensed lead contractor should take the sample and a certified, licensed laboratory should provide the analysis for the TCLP method.

There are three general control strategies for controlling lead paint. These are component replacement, encapsulation, and paint removal. The most cost effective up front control strategy is to encapsulate LBP by repainting areas of chipped, damaged, or deteriorating paint. Cleaning of the component and surface using procedures described above should be completed prior to encapsulation activities. Safe and prompt clean-up of LBP debris or dust when damage or deterioration is encountered should be completed by a licensed lead abatement contractor when possible. LBP component removal and paint removal are considered LBP abatement and would need to be completed by a licensed lead abatement contractor. Should removal or abatement be completed, an exposure assessment should be performed during the work, as required by the OSHA lead-in-construction standard.

FOH recommends personnel responsible for conducting maintenance of LBP components in the facility meet the training requirements for maintenance workers on LBP hazards. This training includes procedures that minimize lead dust generation and soil contamination during maintenance activities, as defined by the EPA and the United States Department HUD. Portions of the OSHA 29 CFR 1910.1025 General Lead Industry Standard apply to maintenance activities that involve making or keeping a structure, fixture, or foundation in proper condition in a routine, scheduled, or anticipated fashion.

6.0 COST ESTIMATE

FOH's estimate for stabilizing the damaged paint by encapsulation with additional paint is approximately 2-3 days of maintenance work.

The programming cost estimate for the removal of lead-based paint or for the demolition of components treated with lead-based paint follows below. In many instances it is a much lower cost to remove, in an intact condition, the treated component and replace it. However, there are some components, which are not practical to remove (i.e. building foundations, etc.) and replace. Demolition costs for such components have not been provided.

This programming cost estimate was derived by use of the Means Cost Guide and by consultation with experienced local contractors. It reflects current industry pricing. Future fluctuations in industry pricing, both natural and seasonal, may affect this estimate. In addition, Service Units outside of metropolitan areas may incur additional costs due to travel requirements on the part of abatement contractors. For purposes of this estimate, an 18% mark-up for logistical costs is included.

This estimate is for abatement contractor costs only. Additional costs that are not included in this estimate are those of an environmental consultant and for replacement of materials that have been removed. Disposal charges have also not been provided. Regulations regarding disposal of LBP-treated items can vary by state. Toxicity Characteristic Leaching Procedure (TCLP) testing may determine that the material may be disposed as general construction waste.

Table 3:
Albuquerque Service Unit
Building No. 21
Cost Estimate

Albuquerque Indian Hospital Building No. 21			**Chemical Removal**		**Component Demolition**	
Room Number	**Building Component**	**Quantity (sf)**	**Unit Cost (sf)**	**Total***	**Unit Cost (sf)**	**Total***
Interior-1930 Section Only	Metal Doorframe Components	2,700	$9.22	$29,374.92	$1.34	$4,269.24
Interior-1930 Section Only	Wood Door Components	120	9.22	1,305.55	1.34	189.74
Interior-1930 Section Only	Wood Doors (X-Ray area only)	135	9.22	1,468.75	1.34	213.46
Interior-1930 Section Only	Metal Window Components	1,350	13.31	21,202.83	5.32	8,474.76
Exterior-1930 Section Only	Metal Window Components	1,350	13.31	21,202.83	5.32	8,474.76
Interior-1930 Section Only	Plaster Walls and Ceilings	70,363	7.98	662,566.15	0.51	42,344.45

Interior-1930 Section Only	Metal Fire Boxes	24	3.73	105.63	0.81	22.94
Exterior	Concrete Curb	200	3.73	880.28	N/A	N/A
Exterior	Asphalt Stripes	500	3.73	2,200.70	N/A	N/A
Interior-Stairwells- 1930 Section Only	Metal Stair Components	2,400	3.73	10,563.36	0.81	2,293.92
Exterior	Metal Fire Hydrant	20	10.10	238.36	N/A	N/A
Room 503	Concrete Walls Side A and C	400	3.73	1,760.56	N/A	N/A
Room 503	Concrete Window Sill	5	3.73	22.01	3.73	22.01
Exterior	Metal Case	30	3.73	132.04	0.81	28.67
Room 119	Brick Wall A and B	410	3.73	1,804.57	N/A	N/A
Rooms 410, 418, 313B, and Hallways	Drywall Walls	6,000	3.99	28,249.20	0.20	1,416
Room S301	Concrete Walls B and C	800	3.73	3,521.12	N/A	N/A
Total*				**$786,598.86**		**$67,749.95**

*Total includes 18% markup.

7.0 PROJECT LIMITATIONS

This Project was performed using, as a minimum, practices consistent with standards acceptable within the industry at this time, and a level of diligence typically exercised by industrial hygiene and environmental consultants performing similar services.

The procedures used in this investigation attempt to establish a balance between the competing goals of limiting investigative and reporting costs and time, and reducing the uncertainty about unknown conditions. Therefore, because the findings of this report were derived from the scope, costs, time, and other limitations, the conclusions should not be construed as a guarantee that all environmental or occupational hazards have been identified and fully evaluated. Where sample collection and testing have been performed, FOH's professional opinions are based in part on the interpretation of data from discrete sampling locations that may not represent conditions at non-sampled locations. FOH assumes no responsibility for omissions or errors resulting from inaccurate information, or data, provided by sources outside of FOH or from omissions or errors in public records.

Furthermore, it is emphasized that the final decision on how much risk to accept always remains with the client since FOH is not in a position to fully understand all of the client's needs. Clients with a greater aversion to risk may want to take additional actions while others, with less aversion to risk, may want to take no further action.

APPENDIX A

Tables of XRF Field Data & Calibrations

Data Table Heading Descriptions

Column Heading	Description
XL No.	Sequential number of each lead measurement
Site	Building number or name
Insp	Inspector initials
Side	Wall (A, B, C, or D) the measurement was taken on
Room	Room number shot was taken in
Sub	Substrate of tested material (wood, concrete, etc.)
Feat	The actual component tested (door, window, etc.)
Cnd	Condition of paint (Intact, Fair, Poor)
Clr	Color of Paint
Result	Negative or Positive for lead content over 1.0mg/cm^2
Pbc±Prec	Result reported in milligrams per centimeter squared (mg/cm^2).

APPENDIX B

Site Map

APPENDIX C

Certifications